

$$\text{Thm 5.1.11} \quad X_\Sigma \simeq (\mathbb{C}^{\Sigma(1)} \setminus Z(\Sigma)) // G$$

$$\text{Let } \widehat{G} = \{ \chi : G \rightarrow \mathbb{C}^* \}$$

$\mathcal{L}_\chi$: linearized line bundle with character χ

$$\begin{array}{ccc} G \curvearrowright \mathcal{L}_\chi \curvearrowright T(\mathbb{C}^r, \mathcal{L}_\chi)^G & \mathcal{L}_\chi \simeq \mathbb{C}^r \times \mathbb{C} & \\ \downarrow \text{Is} & \parallel & \forall (p, t) \in \mathbb{C}^r \times \mathbb{C} \quad g \cdot (p, t) = (g \cdot p, \chi(g)t) \\ \mathcal{O}_{\mathbb{C}^r} & \{ F \in \mathbb{C}[x_1, \dots, x_r] \mid F(g \cdot p) = \chi(g) F(p) \} & \end{array}$$

$$\text{Ex. } X_\Sigma, \quad G = \text{Hom}(\text{Cl}(X_\Sigma), \mathbb{C}^*), \quad \widehat{G} = \text{Cl}(X_\Sigma)$$

$$S = \mathbb{C}[x_\varphi \mid \varphi \in \Sigma(1)] \quad , \quad \beta \in \text{Cl}(X_\Sigma) \curvearrowright \chi \in \widehat{G}$$

$$\Rightarrow T(\mathbb{C}^{\Sigma(1)}, \mathcal{L}_\chi)^G \simeq S_\beta = \{ F \in S \mid \deg(F) = \beta \}$$

Semistable and Stable Points. Given a global section s of $\mathcal{L}_\chi$, note that

$$(\mathbb{C}^r)_s = \{ p \in \mathbb{C}^r \mid s(p) \neq 0 \}$$

is an affine open subset of $\mathbb{C}^r$ since $s(p) = (p, F(p))$ and $s(p) \neq 0$ means $F(p) \neq 0$. Also observe that G acts on $(\mathbb{C}^r)_s$ when s is G -invariant.

Definition 14.1.6. Fix $G \subseteq (\mathbb{C}^*)^r$ and $\chi \in \widehat{G}$, with linearized line bundle $\mathcal{L}_\chi$.

- (a) $p \in \mathbb{C}^r$ is **semistable** if there are $d > 0$ and $s \in \Gamma(\mathbb{C}^r, \mathcal{L}_{\chi^d})^G$ such that $p \in (\mathbb{C}^r)_s$.
- (b) $p \in \mathbb{C}^r$ is **stable** if there are $d > 0$ and $s \in \Gamma(\mathbb{C}^r, \mathcal{L}_{\chi^d})^G$ such that $p \in (\mathbb{C}^r)_s$, the isotropy subgroup G_p is finite, and all G -orbits in $(\mathbb{C}^r)_s$ are closed in $(\mathbb{C}^r)_s$.
- (c) The set of all semistable (resp. stable) points is denoted $(\mathbb{C}^r)_\chi^{\text{ss}}$ (resp. $(\mathbb{C}^r)_\chi^{\text{s}}$).

Proposition 14.1.9. Let X_Σ be a projective toric variety, and let $\Sigma' \subseteq \Sigma$ be the subfan consisting of all simplicial cones of Σ . If $\chi \in \widehat{G}$ comes from an ample divisor class $\beta \in \text{Cl}(X_\Sigma)$, then

$$(\mathbb{C}^{\Sigma(1)})_\chi^{\text{ss}} = \mathbb{C}^{\Sigma(1)} \setminus Z(\Sigma)$$

$$(\mathbb{C}^{\Sigma(1)})_\chi^{\text{s}} = \mathbb{C}^{\Sigma(1)} \setminus Z(\Sigma').$$

GIT Quotient.

$$R_\chi = \bigoplus_{d=0}^{\infty} T(\mathbb{C}^r, \mathcal{I}_{\chi^d}) \subset \mathbb{C}[[x_1, \dots, x_r]]$$

$$\text{Def. } \mathbb{C}^r //_\chi G = \text{Proj}(R_\chi)$$

Proposition 14.1.12. For $G \subseteq (\mathbb{C}^*)^r$ and $\chi \in \widehat{G}$, we have:

- (a) There is a projective morphism $\mathbb{C}^r //_\chi G \rightarrow \mathbb{C}^r // G = \text{Spec}(\mathbb{C}[x_1, \dots, x_r]^G)$.
- (b) $\mathbb{C}^r //_\chi G \neq \emptyset$ if and only if $(\mathbb{C}^r)^\chi \neq \emptyset$.
- (c) The GIT quotient $\mathbb{C}^r //_\chi G$ is a good categorical quotient of $(\mathbb{C}^r)^\chi$ under the action of G , i.e., $\mathbb{C}^r //_\chi G \simeq (\mathbb{C}^r)^\chi // G$.
- (d) If $(\mathbb{C}^r)^\chi \neq \emptyset$, then the action of G on $(\mathbb{C}^r)^\chi$ has a geometric quotient $(\mathbb{C}^r)^\chi / G$ isomorphic to a nonempty open subset of $\mathbb{C}^r //_\chi G$. Thus $\mathbb{C}^r //_\chi G \simeq (\mathbb{C}^r)^\chi // G$ is an almost geometric quotient and $\dim \mathbb{C}^r //_\chi G = r - \dim G$.

$$\begin{aligned} \bullet \quad G &\hookrightarrow (\mathbb{C}^*)^r \xrightarrow{\sim} \mathbb{Z}^r \xrightarrow{\gamma} \widehat{G} \\ 0 &\rightarrow \mathcal{M} \xrightarrow{\delta} \mathbb{Z}^r \xrightarrow{\gamma} \widehat{G} \rightarrow 0 \\ &\quad a \mapsto \gamma(a) = \chi^a \end{aligned}$$

$$\delta : \mathcal{M} \rightarrow \mathbb{Z}^r \xrightarrow{\sim} \mathbb{Z}^r \rightarrow \mathcal{N} \xrightarrow{\sim} \delta \text{ is given by } \begin{aligned} e_i &\mapsto v_i \\ &(\langle m, v_1 \rangle, \dots, \langle m, v_r \rangle) \end{aligned}$$

• The polyhedra of $\chi^a \in \widehat{G}$

I. The first model

$$P_a = \{m \in \mathcal{M} \mid \langle m, v_i \rangle \geq -a_i, 1 \leq i \leq r\}$$

$\subseteq \mathcal{M}_{\mathbb{R}}$

Note: For a fixed $\chi \in \widehat{G}$, a is not unique

P_a is not necessarily lattice polyhedra.

II. The second model

$$0 \rightarrow \mathcal{M}_{\mathbb{R}} \xrightarrow{\delta_{\mathbb{R}}} \mathbb{R}^r \xrightarrow{\gamma_{\mathbb{R}}} \widehat{G}_{\mathbb{R}} \rightarrow 0$$

$$\text{For } \chi \in \widehat{G}, \quad P_\chi = \{b \in \mathbb{R}^r_{\geq 0} \mid \gamma_{\mathbb{R}}(b) = \chi \otimes 1\} \subseteq \mathbb{R}^r$$

$$\text{Note: } P_\chi = \delta_{\mathbb{R}}(P_a) + a$$

$$= \{(\langle m, v_1 \rangle + a_1, \dots, \langle m, v_r \rangle + a_r) \mid m \in P_a\}$$

• GIT quotients are toric

Theorem 14.2.13. For $G \subseteq (\mathbb{C}^*)^r$ and $\chi \in \widehat{G}$, pick $\mathbf{a} \in \mathbb{Z}^r$ such that $\chi = \chi^{\mathbf{a}}$. Then:

- (a) The graded ring R_χ from (14.1.2) satisfies $R_\chi \simeq \mathbb{C}[C(P_{\mathbf{a}}) \cap (M \times \mathbb{Z})]$.
- (b) The GIT quotient $\mathbb{C}^r //_\chi G = \text{Proj}(R_\chi)$ is the toric variety of $P_{\mathbf{a}} \subseteq M_{\mathbb{R}}$.

In Theorem 14.2.13, we used the polyhedron $P_{\mathbf{a}}$ rather than the more intrinsic model P_χ . The reason goes back to Example 14.2.4, where we learned that lattice points in $P_\chi \subseteq \mathbb{R}^r$ relative to $\mathbb{Z}^r$ need not come from lattice points in $P_{\mathbf{a}}$. When $\widehat{G}$ has no torsion (i.e., when G is a torus), this problem goes away and $\mathbb{C}^r //_\chi G$ is the toric variety of the polyhedron $P_\chi \subseteq \mathbb{R}^r$.

Corollary 14.2.16. If $G \subseteq (\mathbb{C}^*)^r$ and $\chi \in \widehat{G}$, then the GIT quotient $\mathbb{C}^r //_\chi G$ is a semiprojective toric variety of dimension

$$\dim \mathbb{C}^r //_\chi G = \dim P_\chi.$$

□

• Semistable point

Let $\chi = \chi^{\mathbf{a}}$, the polyhedra $P_\chi = P_{\mathbf{a}}$
 Let $F_{i,\mathbf{a}} = \{m \in P_{\mathbf{a}} \mid \langle m, v_i \rangle = -a_i\}$
 $F_{i,\chi} = P_\chi \cap V(x_i)$

Proposition 14.2.21. Let $p = (p_1, \dots, p_r) \in \mathbb{C}^r$ and set $I(p) = \{i \mid 1 \leq i \leq r, p_i = 0\}$. Then $p \in (\mathbb{C}^r)_\chi^{\text{ss}}$ if and only if $\bigcap_{i \in I(p)} F_{i,\chi} \neq \emptyset$.

pf. " $\Rightarrow$ " If p is s.s., then $\exists (G, \chi^d)$ -invariant F s.t. $F(p) \neq 0$

$\Rightarrow \exists (G, \chi^d)$ -invariant monomial x^b of F s.t. $x^b(p) \neq 0$

$\Rightarrow b \in \mathcal{D}P_\chi \rightarrow \left(\begin{array}{l} x^b \text{ is } (G, \chi^d)\text{-invariant} \\ \Leftrightarrow r(b) = \chi^d \\ \Rightarrow b \in P_{\chi^d} = \mathcal{D}P_\chi \end{array} \right.$

Since $x^b(p) \neq 0$

$\Rightarrow b_i = 0$ for all $i \in I(p)$

$\Rightarrow b \in \bigcap_{i \in I(p)} \mathcal{D}F_{i,\chi} \Rightarrow \bigcap_{i \in I(p)} F_{i,\chi} \neq \emptyset$ $x = 1$
 $\chi^b(g) = x^b(g) = \chi^d(g)$

$$" \in " \cap_{i \in I(p)} F_{i,\chi} \neq \emptyset \Rightarrow \exists d \text{ s.t. } b \in \left(\bigcap_{i \in I(p)} d F_{i,\chi} \right) \cap \mathbb{Z}^r$$

$$\Rightarrow \textcircled{1} \chi^b(p) \neq 0$$

$$\textcircled{2} r_{\mathbb{R}}(b) = \chi^d \otimes 1, \text{ so that } r(b) \text{ and } \chi^d$$

$$r(b) = \chi^b, \chi^b - \chi^d = q \text{ differ by a torsion element of } \hat{A}.$$

$$\Rightarrow \exists \text{ integer } l > 0, \chi^{lb} \text{ is } (h, \chi^{ld})\text{-invariant}$$

$$r(lb) = \chi^{lb} \quad \chi^{lb}(p) \neq 0 \Rightarrow p \text{ is s.s.}$$

$$\exists l \text{ s.t. } \chi^{lb} = (q + \chi^d)l = \chi^{ld}$$

□

When we think about semistable points algebraically, we get the ideal

$$(14.2.11) \quad B(\chi) = \langle \prod_{i \notin I} x_i \mid I \subseteq \{1, \dots, r\}, \bigcap_{i \in I} F_{i,\chi} \neq \emptyset \rangle \subseteq S = \mathbb{C}[x_1, \dots, x_r],$$

called the *irrelevant ideal* of χ . The vanishing locus of $B(\chi)$ is the *exceptional set* $Z(\chi) = \mathbf{V}(B(\chi)) \subseteq \mathbb{C}^r$. This gives a nice description of the set of semistable points, which you will prove in Exercise 14.2.9.

Corollary 14.2.22. $(\mathbb{C}^r)_{\chi}^{\text{ss}} = \mathbb{C}^r \setminus Z(\chi)$, so that $\mathbb{C}^r //_{\chi} G \simeq (\mathbb{C}^r \setminus Z(\chi)) // G$. □

$$\text{pf. } p \in (\mathbb{C}^r)_{\chi}^{\text{ss}} \Leftrightarrow \bigcap_{i \in I(p)} F_{i,\chi} \neq \emptyset$$

$$\Leftrightarrow \prod_{i \notin I(p)} x_i \in B(\chi)$$

$$\Leftrightarrow p \notin Z(\chi)$$

□

• The two cones

$$0 \rightarrow M \xrightarrow{\delta} \mathbb{Z}^r \xrightarrow{\tau} \hat{G} \rightarrow 0$$

$$\mathbb{Z}^r \xrightarrow{\delta^*} N$$

$$\leadsto 0 \rightarrow M_{\mathbb{R}} \xrightarrow{\delta_{\mathbb{R}}} \mathbb{R}^r \xrightarrow{\tau_{\mathbb{R}}} \hat{G}_{\mathbb{R}} \rightarrow 0 \quad \leadsto \quad \mathbb{R}^r \xrightarrow{\delta_{\mathbb{R}}^*} N_{\mathbb{R}}$$

$$e_i \mapsto \beta_i \quad e_i \mapsto v_i$$

$$m \mapsto (\langle m, v_1 \rangle, \dots, \langle m, v_r \rangle)$$

$$C_{\beta} = \text{Cone}(\beta) = \text{Cone}(\beta_1, \dots, \beta_r) \subseteq \hat{G}_{\mathbb{R}}$$

$$C_{\nu} = \text{Cone}(\nu) = \text{Cone}(v_1, \dots, v_r) \subseteq N_{\mathbb{R}}$$

$$\beta_i = " [D_i] \Rightarrow C_{\beta}: \text{cone of effective } \mathbb{R}\text{-divisor}$$

$$v_i = " v_i \Rightarrow |C_{\nu}| = |\Sigma|$$

• Existence of semistable and stable point

Proposition 14.3.5. If $\chi \in \hat{G}$ is a character, then the following are equivalent:

- (a) $\mathbb{C}^r //_{\chi} G \neq \emptyset$.
- (b) $(\mathbb{C}^r)_{\chi}^{\text{ss}} \neq \emptyset$.
- (c) $\chi \otimes 1 \in C_{\beta}$.

Proposition 14.3.6. If $\chi \otimes 1 \in C_{\beta}$, then the following are equivalent:

- (a) $(\mathbb{C}^*)^r \subseteq (\mathbb{C}^r)_{\chi}^s$.
- (b) $(\mathbb{C}^r)_{\chi}^s \neq \emptyset$.
- (c) $\chi \otimes 1$ is contained in the interior of C_{β} .
- (d) $F_{i,\chi}$ is a proper face of P_{χ} for all i .

Furthermore, the above conditions imply that

- (e) $\dim \mathbb{C}^r //_{\chi} G = r - \dim G$,

and when ν consists of nonzero vectors, (e) is equivalent to (a)–(d).

Proposition 14.3.6 has many nice consequences concerning $\mathbb{C}^r //_{\chi} G$ when ν consists of nonzero vectors, including:

- Stable points exist if and only if the GIT quotient has the expected dimension.
- The GIT quotient is nonempty of smaller than expected dimension if and only if the character is on the boundary of C_{β} .

Ex Hirzebruch surface $\mathcal{H}_r$ $r=1$

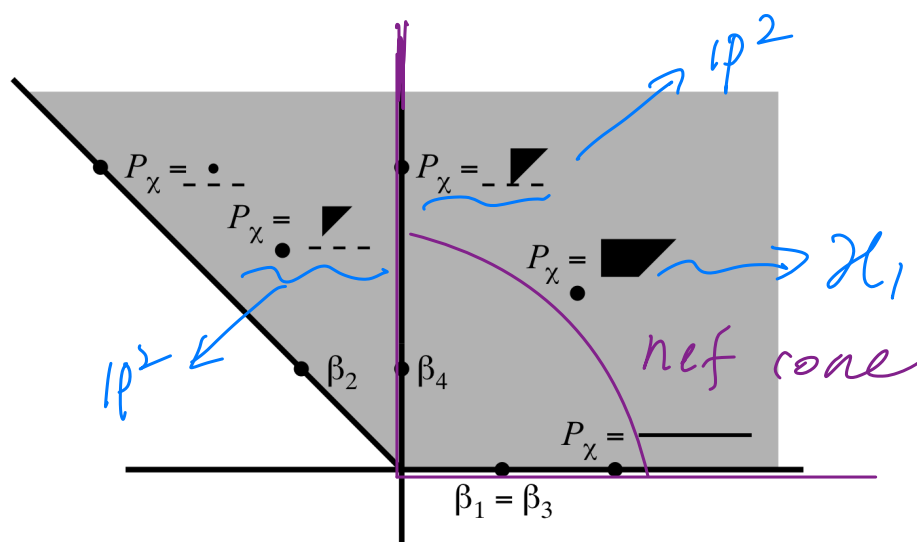


Figure 5. C_β and the polytopes P_χ for all $\chi \otimes 1 \in C_\beta$

generic character

Def. $\chi \in \hat{G}$ is generic if $\chi \otimes 1 \in C_\beta$ and for every $\beta' \subseteq \beta$ with $\dim \text{Cone}(\beta') < \dim \text{Cone}(\beta) = \dim G$ we have $\chi \otimes 1 \notin \text{Cone}(\beta')$

Theorem 14.3.14. For $\chi \otimes 1 \in C_\beta$, the following are equivalent:

- (a) χ is generic.
- (b) Every vertex of P_χ has precisely $\dim G$ nonzero coordinates.
- (c) P_χ is simple of dimension $r - \dim G$, every virtual facet $F_{i,\chi} \subseteq P_\chi$ is either empty or a genuine facet, and $F_{i,\chi} \neq F_{j,\chi}$ if $i \neq j$ and $F_{i,\chi}, F_{j,\chi}$ are nonempty.
- (d) $(\mathbb{C}^r)_\chi^s = (\mathbb{C}^r)_\chi^{ss}$.

• Projective and birational GIT Quotients

We have $\mathbb{C}^r //_\chi G \rightarrow \mathbb{C}^r // G = \text{Spec}[\mathbb{C}[x_1, \dots, x_r]^G]$ is projective

• Two extreme case where $\mathbb{C}^r // G$ is very small or very large.

1. very small

Proposition 14.3.10. *The following are equivalent:*

- (a) $\mathbb{C}^r //_{\chi} G$ is projective for all $\chi \otimes 1 \in C_{\beta}$.
- (b) $\mathbb{C}^r //_{\chi} G$ is projective for some $\chi \otimes 1 \in C_{\beta}$.
- (c) $\mathbb{C}[x_1, \dots, x_r]^G = \mathbb{C}$.
- (d) β consists of nonzero vectors and C_{β} is strongly convex.

2. very large

Proposition 14.3.11. *The following conditions are equivalent:*

- (a) $\dim \mathbb{C}^r // G = r - \dim G$.
- (b) $C_{\nu} \subseteq N_{\mathbb{R}}$ is strongly convex.
- (c) The map $\mathbb{C}^r //_{\chi} G \rightarrow \mathbb{C}^r // G$ from (14.3.6) is birational for all $\chi \otimes 1 \in C_{\beta}$.

Furthermore, when ν consists of nonzero vectors, we can replace (b) and (c) with:

- (b') $C_{\beta} = \widehat{G}_{\mathbb{R}}$.
- (c') $\mathbb{C}^r //_{\chi} G \rightarrow \mathbb{C}^r // G$ is birational for all χ .

Ex.

Example 14.3.12. For $G = \{(t, t, t^{-1}) \mid t \in \mathbb{C}^*\}$, one gets $\beta_1 = \beta_2 = e_1, \beta_3 = -e_1$ in $\widehat{G}_{\mathbb{R}} \simeq \mathbb{R}$ and $\nu_1 = e_1, \nu_2 = e_2, \nu_3 = e_1 + e_2$ in $N_{\mathbb{R}} \simeq \mathbb{R}^2$. Thus $C_{\nu} = \text{Cone}(e_1, e_2)$ is strongly convex and $\nu_i \neq 0$ for all i . Hence all conditions of Proposition 14.3.11 apply, i.e., for all $\chi \in \widehat{G}$, we have a birational morphism

$$\mathbb{C}^3 //_{\chi} G \rightarrow \mathbb{C}^3 // G = \text{Spec}(\mathbb{C}[xz, yz]) = \mathbb{C}^2.$$

To understand $\mathbb{C}^r //_{\chi} G$, consider Figure 6, which shows the polyhedron P_{χ} for all χ . For “positive” χ , $\mathbb{C}^r //_{\chi} G$ is the blowup of $\mathbb{C}^2$ at the origin, while for

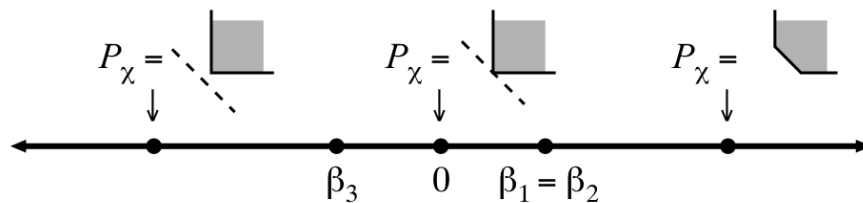


Figure 6. The polytopes P_{χ} for all χ

“negative” χ , the GIT quotient is $\mathbb{C}^2$. The dashed lines in the figure indicate the presence of a virtual facet, which is empty for negative χ and consists of the vertex when χ is the origin. In Exercise 14.3.4 you will compare this to the quotient construction of $\text{Bl}_0(\mathbb{C}^2)$. $\diamond$

§14.4 Secondary fan.

Given $a \in \mathbb{Z}^r$, the polyhedron $P_a \subseteq M_{\mathbb{R}}$ gives

(1) The generalized fan Σ , which is the normal fan of P_a

(2) $I_\phi = \{i \mid F_{i,a} = \emptyset\}$

Then

Proposition 14.4.1. Given a generalized fan Σ in $N_{\mathbb{R}}$ and subset $I_\emptyset \subseteq \{1, \dots, r\}$, then Σ and $I_\emptyset$ come from some $a \in \mathbb{Z}^r$ if and only if

(a) $|\Sigma| = C_\nu$.

(b) X_Σ is semiprojective.

(c) $\sigma = \text{Cone}(\nu_i \mid \nu_i \in \sigma, i \notin I_\emptyset)$ for every $\sigma \in \Sigma$.

• GKZ cones and secondary fan

• Let (Σ, I_ϕ) come from some $a \in \mathbb{Z}^r$. Then the GKZ cone of (Σ, I_ϕ) is

$$\overline{\Gamma}_{\Sigma, I_\phi} = \{(a_1, \dots, a_r) \in \mathbb{R}^r \mid \text{there is } \varphi \in \text{CSF}(\Sigma) \text{ s.t. } \varphi(\nu_i) = -a_i \text{ for } i \notin I_\phi \text{ and } \varphi(\nu_i) \geq -a_i \text{ for } i \in I_\phi\}$$

when $a \in \overline{\Gamma}_{\Sigma, I_\phi}$, φ is unique, denoted by $\varphi = \rho_a$

$$\overline{\Sigma}_{\text{GKZ}} = \{\overline{\Gamma}_{\Sigma, I_\phi} \mid (\Sigma, I_\phi) \text{ comes from } a \in \mathbb{Z}^r\}$$

↓

It is a generalized fan (Lemma 14.4.3 - Lemma 14.4.6)
and $|\overline{\Sigma}_{\text{GKZ}}| = \partial \mathbb{R}^r(C_\beta)$

• For $\overline{\Sigma}_{\text{GKZ}}$, the minimal face is $\ker(\partial_{\mathbb{R}})$

$$\Rightarrow \overline{\Gamma}_{\Sigma, I_\phi} = \overline{\Gamma}_{\Sigma, I_\phi} / \ker(\partial_{\mathbb{R}}) \subseteq \mathbb{R}^r / \ker(\partial_{\mathbb{R}}) = \widehat{G}_{\mathbb{R}}$$

$$\Sigma_{\text{GKZ}} = \{\overline{\Gamma}_{\Sigma, I_\phi} \mid (\Sigma, I_\phi) \text{ comes from } a \in \mathbb{Z}^r\}$$

$$|\Sigma_{\text{GKZ}}| = C_\beta \quad \dim(C_\beta) = \dim G$$

The maximal cones of Σ_{GKZ} are called the chambers of secondary fan

Theorem 14.4.7.

- (a) Σ_{GKZ} is a fan in $\widehat{G}_{\mathbb{R}}$ with $|\Sigma_{\text{GKZ}}| = C_{\beta}$.
- (b) $\Gamma_{\Sigma', I'_{\emptyset}} \preceq \Gamma_{\Sigma, I_{\emptyset}}$ if and only if Σ refines Σ' and $I'_{\emptyset} \subseteq I_{\emptyset}$.
- (c) $\mathbb{C}^r //_{\chi} G \simeq X_{\Sigma}$ when $\chi \otimes 1 \in \text{Relint}(\Gamma_{\Sigma, I_{\emptyset}})$.

Remark: the GIT quotient is constant on the relative interiors of the GKZ cones.